

```

#version 460
#extension GL_EXT_shader_16bit_storage : require
#extension GL_KHR_shader_subgroup_basic : require
#extension GL_KHR_shader_subgroup_arithmetic : require

layout(local_size_x = 32, local_size_y = 1, local_size_z = 1) in;

layout(set = 0, binding = 0) readonly buffer InputFP16_W32 {
    float16_t in_data16_w32[];
};
layout(set = 0, binding = 1) writeonly buffer OutputFP16_W32 {
    float16_t out_data16_w32[];
};

void main() {
    uint idx = gl_GlobalInvocationID.x;

    // 순수 FP16 단독 연산 패스
    float16_t val = in_data16_w32[idx];
    float16_t res = val * val + float16_t(2.0);

    // 16비트 타입 서브그룹 연산
    out_data16_w32[idx] = subgroupAdd(res);
}

```